

HOC

v/s

Custom Hook

What is a Higher Order Component?

A Higher Order Component (HOC) is a function that takes a component and returns a new component with enhanced or additional behavior.

Component → HOC → Enhanced Component.

It follows the same idea as higher-order functions in JavaScript.

Basic Syntax of a HOC

```
const withRedBackground = (WrappedComponent) => {  
  return (props) => {  
    return (  
      <div style={{ backgroundColor: "red", padding: "20px" }}>  
        <WrappedComponent {...props} />  
      </div>  
    );  
  };  
};
```

```
const EnhancedComponent = withBlueBackground(MyComponent);
```

What is a Custom Hook?

custom hook is a JavaScript function in React that:

- ❑ starts with the word use
- ❑ uses other hooks inside it
- ❑ allows you to reuse logic across components

Custom hooks do NOT return JSX. They return data or functions. They help extract reusable logic, not UI.

Why do we use HOCs Or Custom Hook?

HOCs are used to reuse logic across multiple components without repeating code.

Common Use Cases:

- ☐ Adding extra UI behavior
- ☐ Authentication checks
- ☐ Permission-based rendering
- ☐ Data fetching
- ☐ Logging or analytics
- ☐ Injecting extra props
- ☐ Reusing complex logic across multiple components

HOCs vs Custom Hooks (Modern React)

Nowadays, people prefer custom hooks, because they are:

- ☐ Cleaner
- ☐ Easier to reuse
- ☐ Avoid wrapper hell

But HOCs are still used in many old codebases and libraries like react-redux (connect HOC).

**Thank
You**